

DeepJoint: Robust Survival Modelling Under Clinical Presence Shift

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Problem

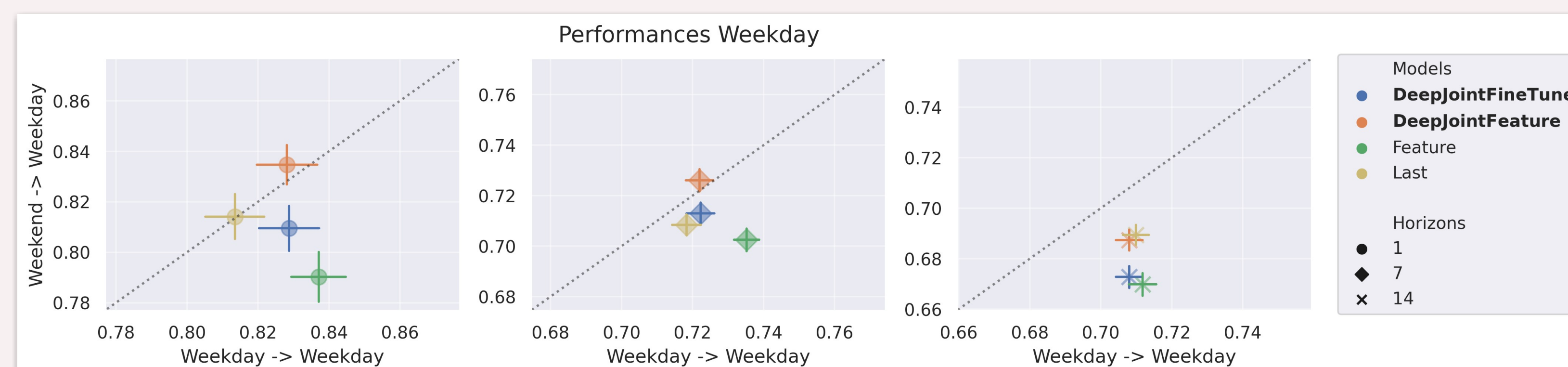
Clinical data arise from an **informative** sampling process, when, which, and how often tests are ordered, yet survival models treat this **clinical presence** as noise, hurting both accuracy and transportability when practice evolves, discarding the very signal that should inform prediction.

Motivation

- The same patients present differently under different observation processes; a model overfit to one regime fails when practice shifts.
- The **weekend effect**, differing test intensity and mortality for weekend vs weekday admissions, offers a controllable shift to stress-test robustness.
- Prior work uses missingness or timing *in isolation*, not the full observation process jointly with survival.

Clinical presence is **signal to model**, not noise to ignore.

Robustness to Shift



Transferred (y-axis) vs oracle (x-axis) discrimination under the weekday $\leftrightarrow$ weekend shift, at 1 / 7 / 14-day horizons. Closer to the diagonal = more robust.

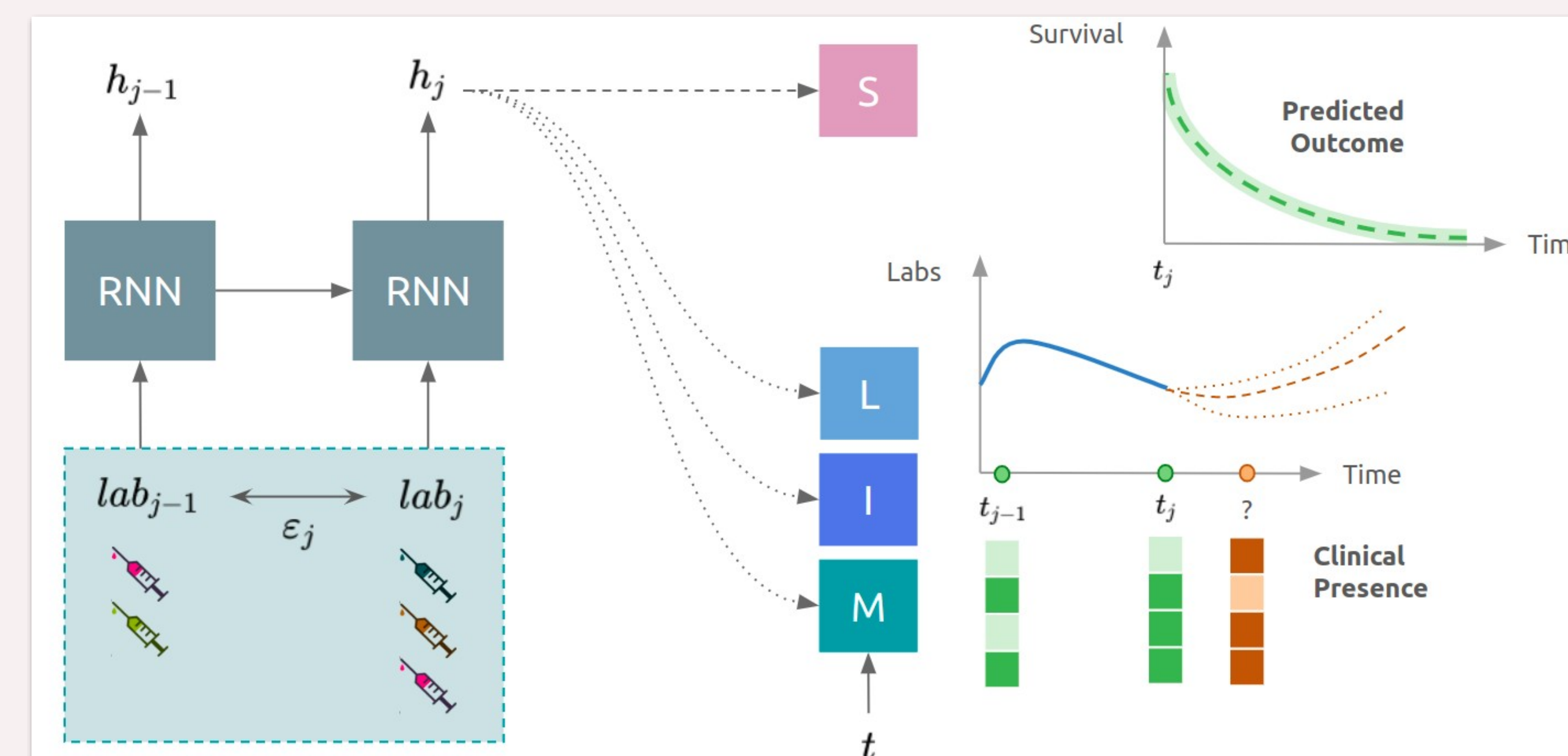
DeepJointFeature stays closest to the diagonal, most robust to the shift; the fine-tuned variant overfits and *Feature* alone is shift-sensitive.

Method

- An **LSTM** encodes each patient's irregular ICU test sequence into a shared embedding h ; three heads then model clinical presence, longitudinal values (*Gaussian*), missingness (*Bernoulli*) and timing (*cumulative intensity*), while a **DeepSurv** Cox head models survival, all trained jointly.

$$\ell(e) = (1 - \alpha) \ell_S + \alpha \sum_{\text{task} \in \{L, I, M\}} w_{\text{task}}(e) \ell_{\text{task}}(e)$$

Joint loss: survival term ℓ_S plus dynamically-weighted clinical-presence tasks; α balances the two.



Deep Joint Model: an LSTM embedding h drives the longitudinal (L), inter-observation (I), missingness (M) and survival (S) networks.

Headline Numbers

0.878

C-Index · 1-day horizon · MIMIC-III
State-of-the-art vs input-matched models

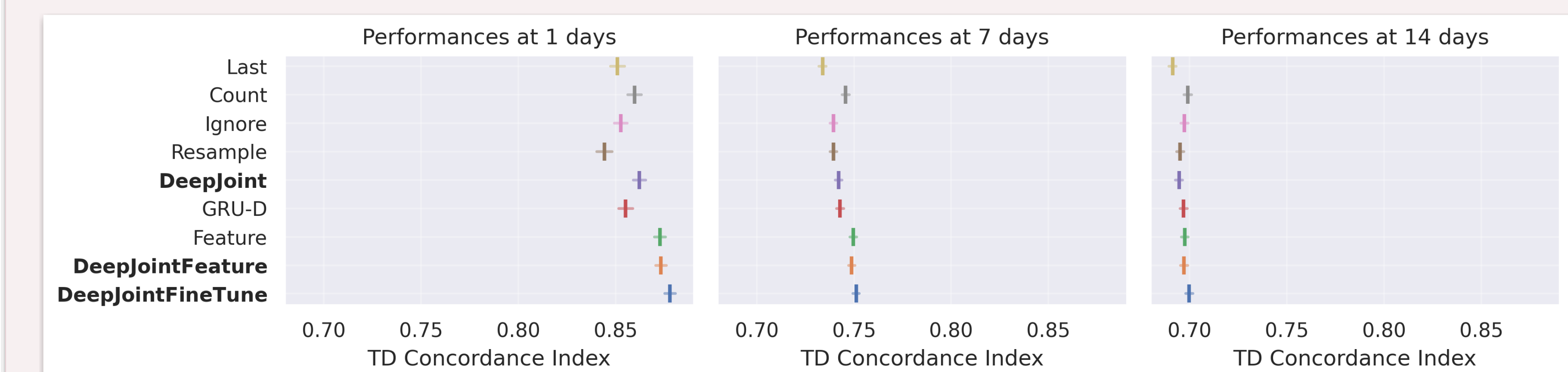
30,834 ICU patients

3 presence processes

1/7/14 day horizons

Key Results

Even lab-only DeepJoint beats GRU-D and the “Ignore” LSTM.



Time-dependent C-Index at 1 / 7 / 14-day horizons (higher is better).
Proposed models (bold) lead at short horizons.

SO WHAT → The edge fades at longer horizons as presence signals short-term instability.

Data & Cohort

- MIMIC-III** ICU laboratory tests, 38,597 patients at Beth Israel Deaconess (2001–12).
- Filtered to **30,834** patients surviving the first 24 h post-admission.

30,834
patients

90/10
train / test split

1·7·14
day horizons

Takeaway

Explicitly modelling **how** clinical data are observed, not just their values, yields survival models that are both more accurate and far more **transportable** when clinical practice shifts. Clinical presence is signal, not noise, and building it into the model is what makes medical predictions transfer across settings, and even flags when the observation process itself has changed.